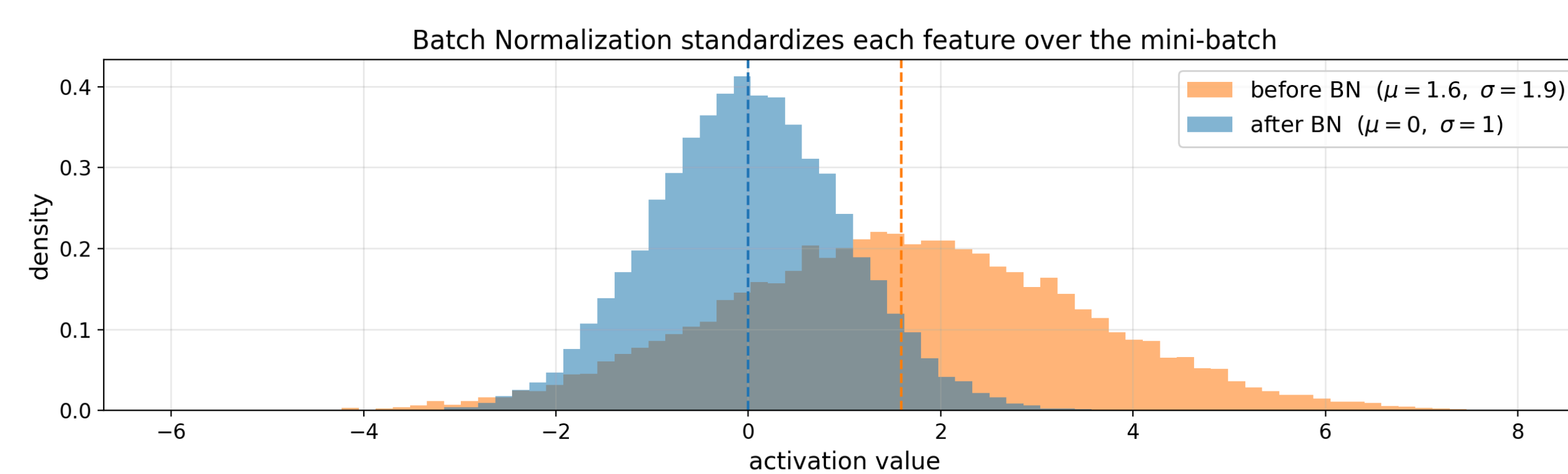


01. Introduction

Batch Normalization (BN) rescales every feature of a mini-batch to mean 0 and variance 1:

$$y = \gamma \cdot (x - \mu_B) / \sigma_B + \beta$$

This simple step lets networks train faster and reach **higher accuracy**, making BN a standard layer in deep learning.



02. Objective

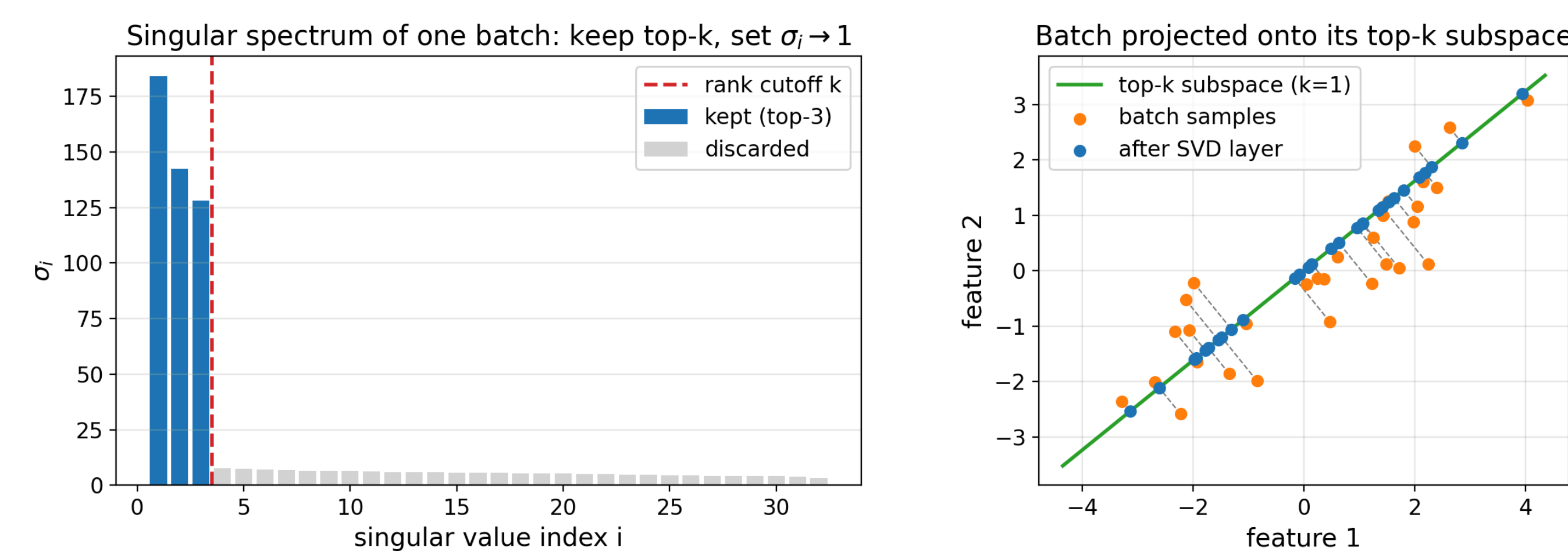
Rank Reduction Autoencoders (Mounayer et al., 2024) regularize networks with the truncated **SVD**, no extra loss needed. Like BN, it **changes the statistics of the batch**.

Can the SVD alone normalize a batch as well as BN?
We build it into a standalone **SVD layer** to find out.

03. Methodology

The SVD layer keeps the batch's **top-k singular directions** and normalizes them by their singular values:

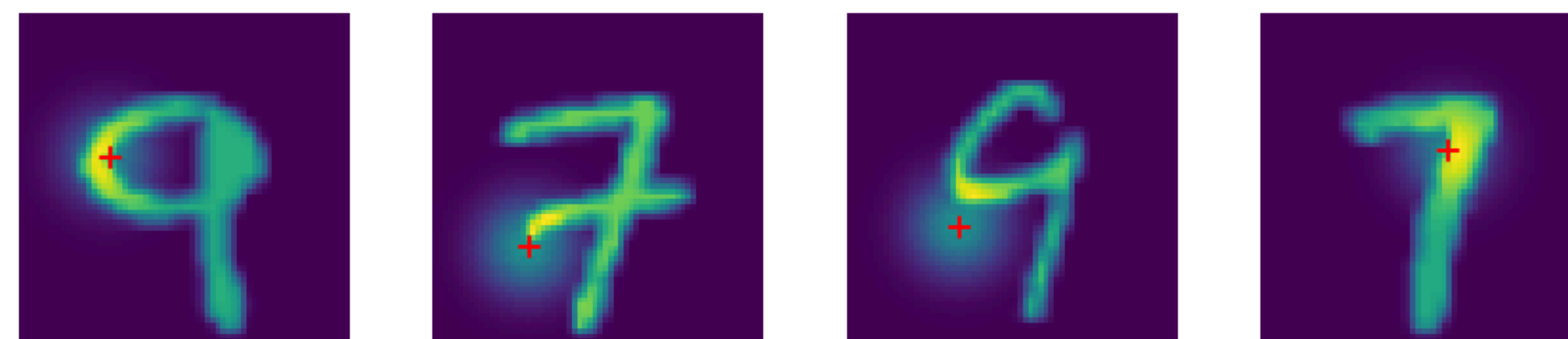
$$X = U \Sigma V^T \rightarrow \hat{X} = \sqrt{N} \cdot U_k V_k^T \quad (\sigma_i \rightarrow 1)$$



BN-style normalization, but in the batch's own decorrelated SVD basis.

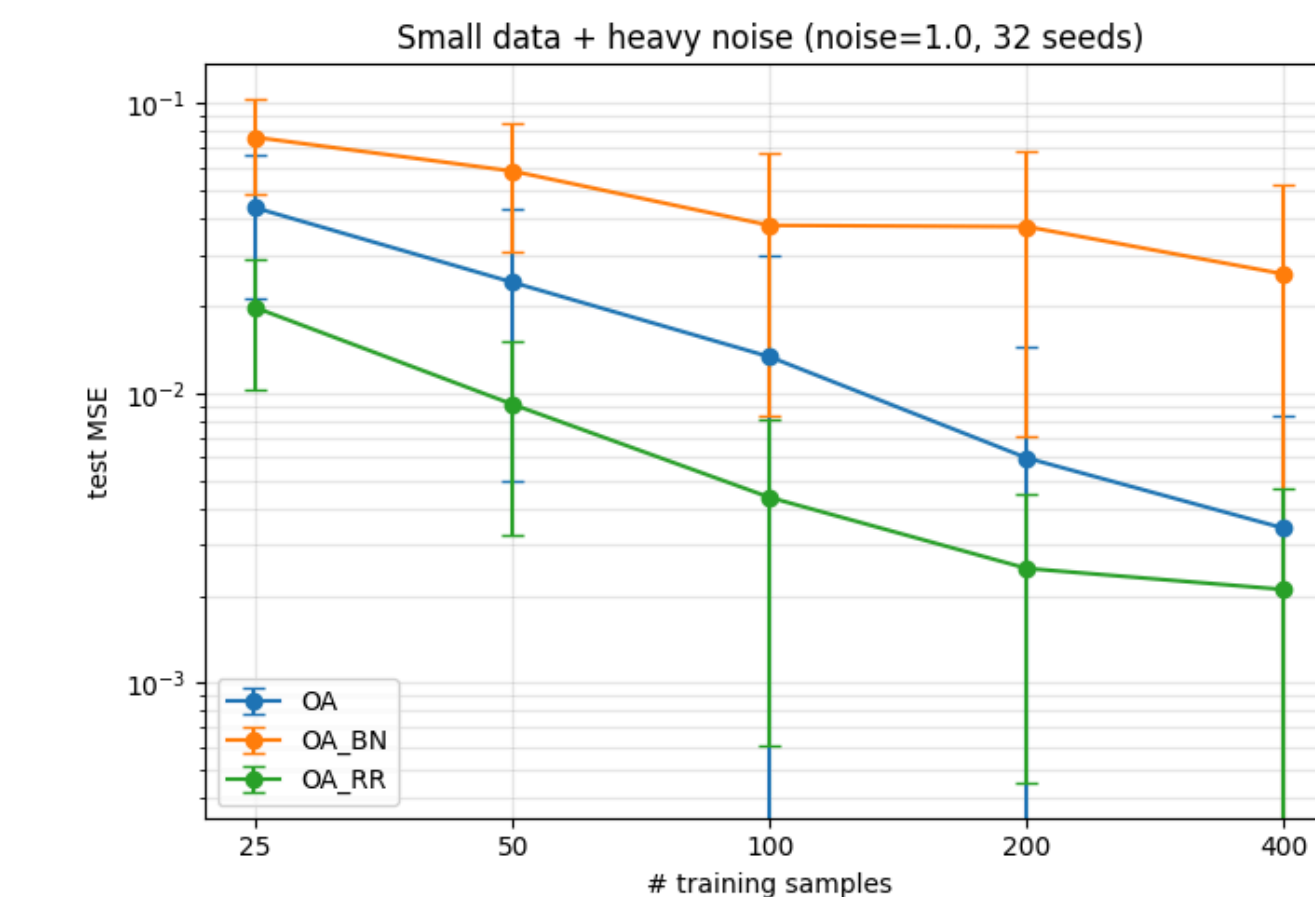
- Task: regress a Gaussian source's centre (x, y) from a 64×64 image: **point-source localization**, as in astrometry and microscopy.
- Compare **Plain vs BN vs SVD** from **identical initial weights**, over up to 32 seeds, sweeping rank k, noise level and type, and dataset size N.
- External validity via **source injection**: a known source planted on real image backgrounds (astronomy's own validation method).

Injected source (red +) on real MNIST backgrounds

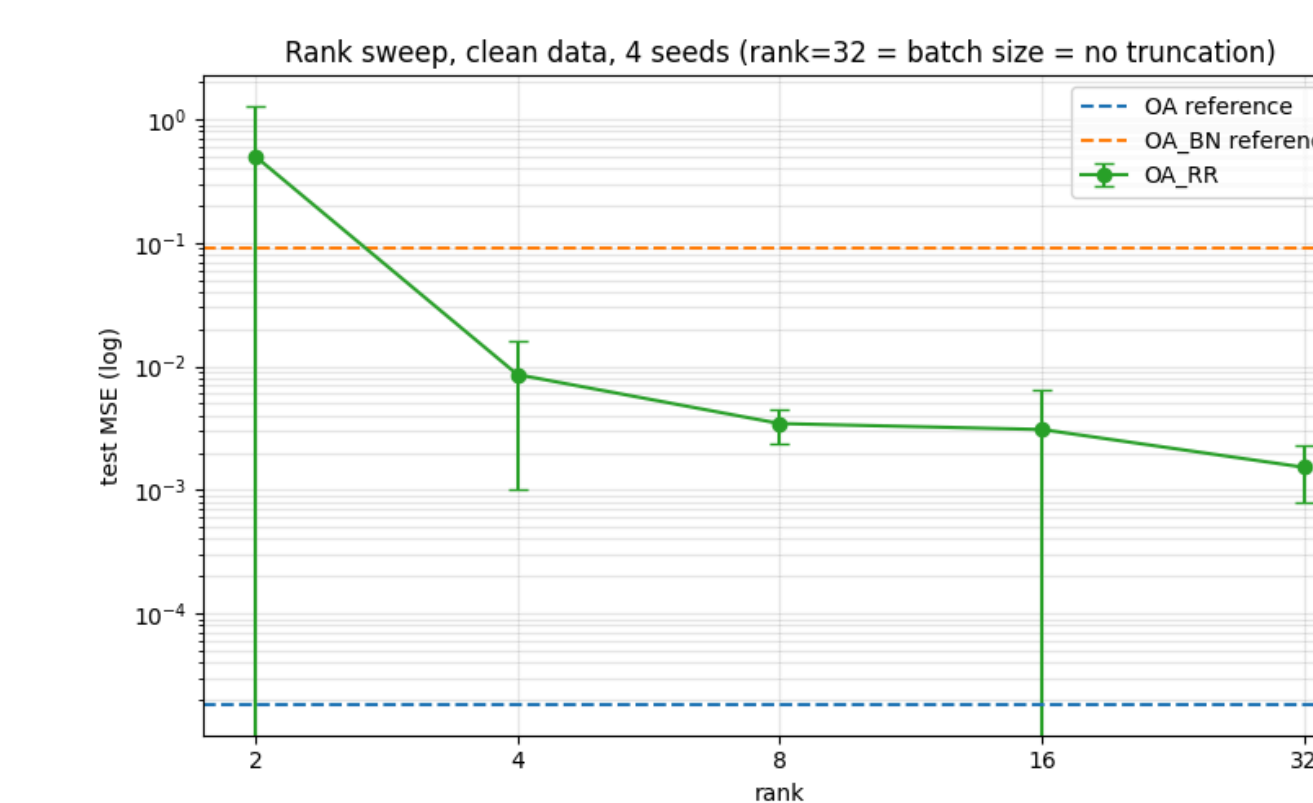


Injected source (red +) on real backgrounds: ground truth stays exact, clutter statistics become real.

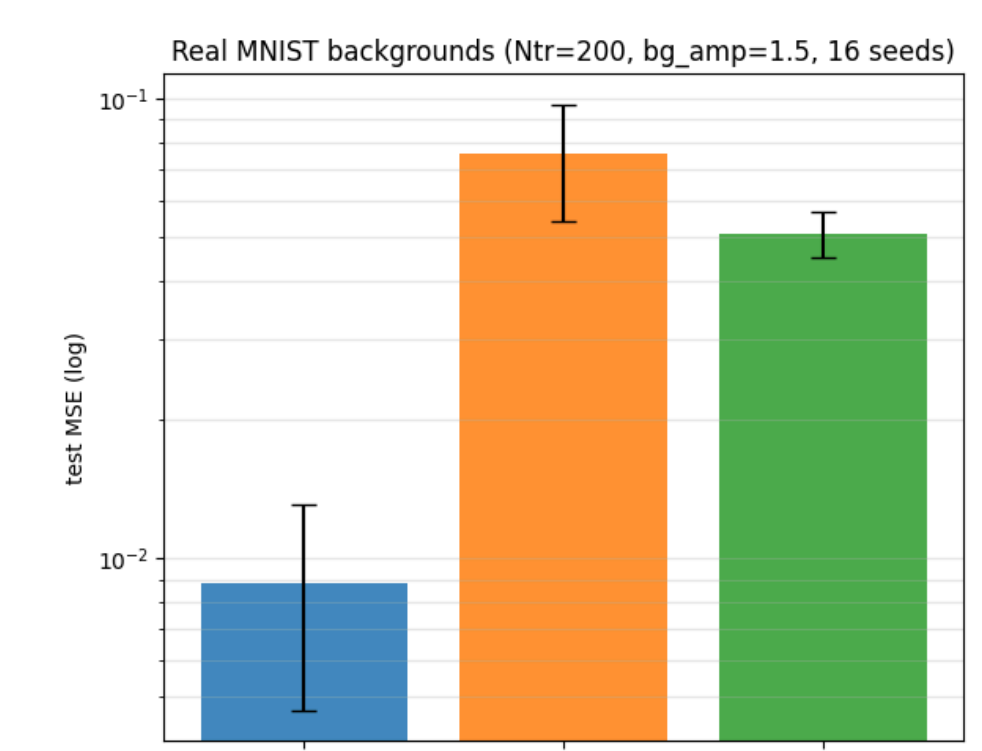
04. Results



Heavy input noise, 32 seeds: the per-layer SVD layer (green) beats BN *and* the plain net at every training-set size. (OA = plain, OA_RR = SVD layer.)



Error falls monotonically with rank k (bottleneck ablation).



Real backgrounds: plain wins; SVD still beats BN.

- SVD beats BN in every comparison we ran** (clean, noisy, small-data, real backgrounds), by 1.5–15×.
- Under noise it beats **even the unregularized net at every dataset size** (2–3×), with the tightest seed variance and near-zero train/test gap.
- An earlier per-layer instability traced to a custom analytic SVD backward. **Replaced by native autograd**, per-layer placement wins.

05. Conclusion

- The SVD layer is a **more robust drop-in alternative to BN**, and under input noise outperforms even the unregularized network at every dataset size we tested.
- Its one weakness: keeping top-k passes **structured low-rank nuisance** through, the inverse of astronomy's KLIP. A **reject-top-k** variant is the natural next step.

References

- J. Mounayer et al. *Rank Reduction Autoencoders*. arXiv:2405.13980, 2024.
- S. Ioffe, C. Szegedy. *Batch Normalization*. ICML 2015.
- R. Soummer et al. *Detection and Characterization of Exoplanets using Projections on Karhunen-Loève Eigenimages (KLIP)*. ApJL 2012.